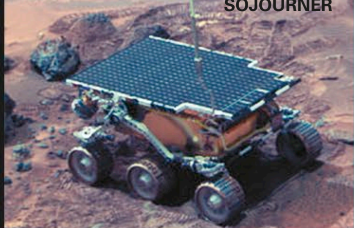
MISSIONS TO MARS

Numerous spacecraft have been sent to Mars since the first missions were undertaken by the USA and the Soviet Union in the 1960s, with varying success due to technical difficulties. A selection of successful missions is described below.

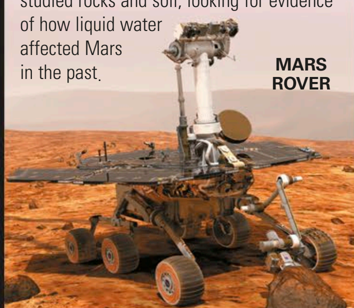
1976 VIKING 1 AND 2 (USA) These two craft each consisted of an orbiter and a lander. The orbiters sent back images, while the landers descended to two different sites and sent back analyses of the soil and atmosphere, as well as images.

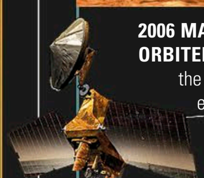
1997 MARS PATHFINDER (USA) his mission sent a stationary lander and a free-ranging robot called Sojourner to the surface of Mars. They landed in an ancient floodplain and sent back pictures and analyses of soil samples.


SOJOURNER

2003 MARS EXPRESS (EUROPE) This orbiting spacecraft is imaging the entire surface of Mars, as well as mapping its mineral composition and studying the Martian atmosphere.

2004 MARS EXPLORATION ROVERS (USA) Twin rovers Spirit and Opportunity landed on opposite sides of the planet and studied rocks and soil, looking for evidence of how liquid water affected Mars in the past.


MARS ROVER


MRO

2006 MARS RECONNAISSANCE ORBITER (USA) Looping over the poles of Mars 12 times every Martian day, MRO keeps a constant eye on the red planet's weather and looks for signs of water, past or present, on its surface.

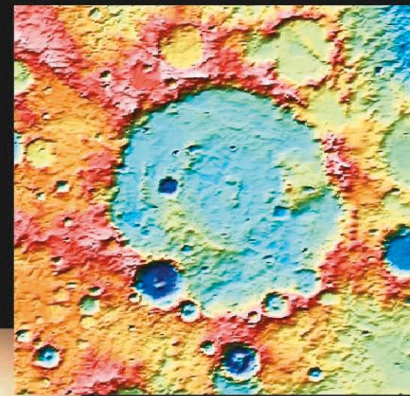
2012 CURIOSITY (USA) About the size of a car, this rover is exploring Gale Crater.

2016 EXOMARS TRACE GAS ORBITER (EUROPE AND RUSSIA) This orbiter searches for methane and other minor atmospheric components.

2018 INSIGHT (USA) A stationary lander, InSight studies marsquakes and heat flow.

SURFACE FEATURES

Mars's surface features have been formed and shaped by meteorite impacts, by the wind (see p.151), and by volcanism and faulting (see Tectonic Features, below). Scientists also believe that water once flowed on and below the surface of Mars (see opposite), carving out features such as valleys and outflow channels. The craters formed during a period of intense meteorite bombardment about 3.9 billion years ago. They are found mainly in the southern hemisphere, which is geologically older than the northern hemisphere, and include the vast Hellas Basin (see p.165), but small craters are found all over Mars. Martian craters are flatter than those on the Moon and show signs of erosion by wind and water; indeed, some have almost been obliterated.


IMPACT CRATER
The Herschel impact crater, located in the southern highlands, is about 185 miles (300 km) across. This image has been false-colored to show altitude. The lowest areas are the dark blue floors of smaller craters. The Herschel Crater floor is mostly at 3,240 ft (1,000 m) and the highest parts of the rim (pale pink) are at about 9,720 ft (3,000 m).

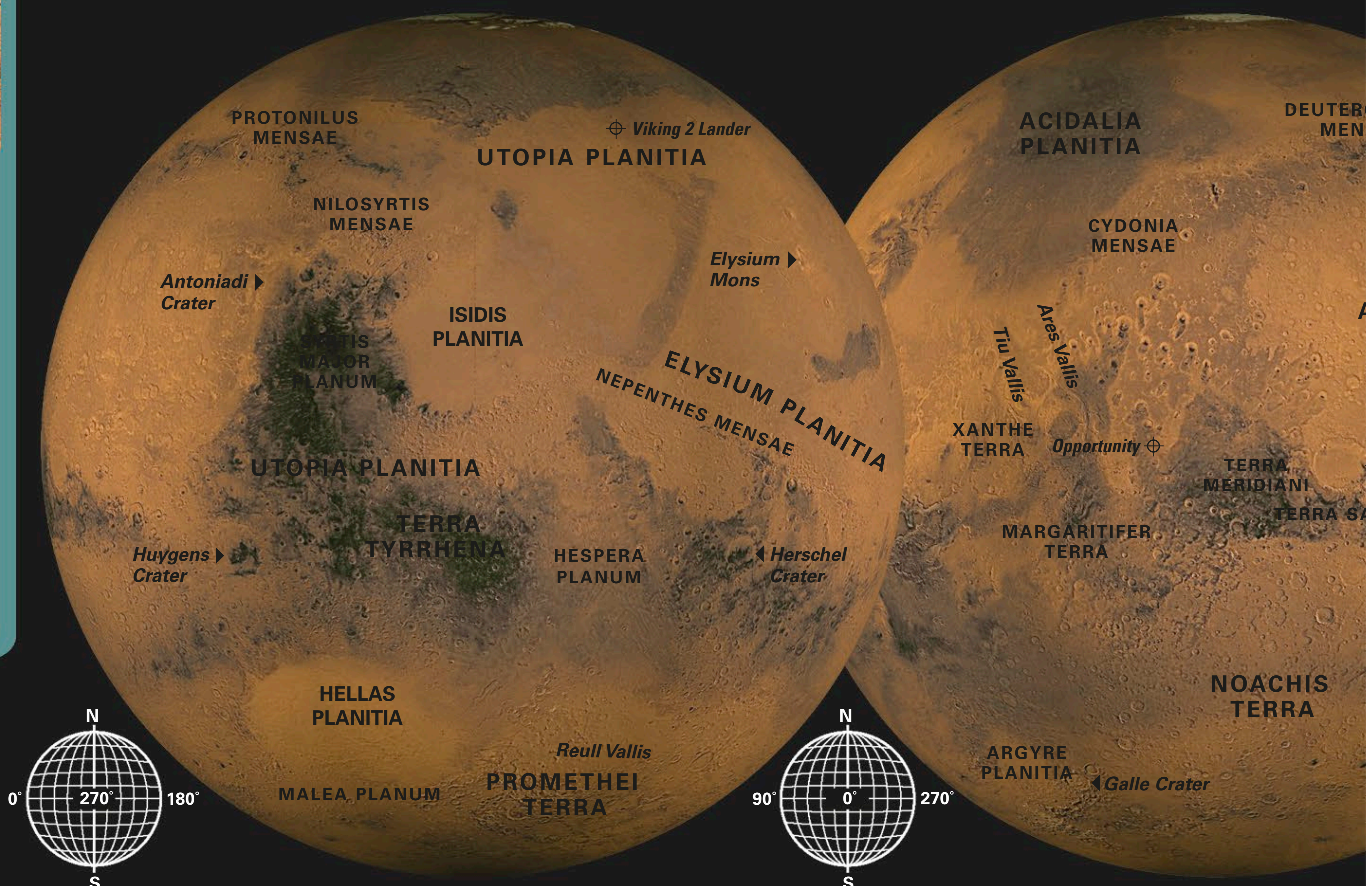

SPIRIT AT HUSBAND HILL
The Spirit rover's arm reaches out to investigate a rocky outcrop called Hillary, named after the mountaineer Sir Edmund Hillary, near the summit of Husband Hill.

TECTONIC FEATURES

Billions of years ago, when Mars was a young planet, internal adjustments created the large-scale features seen on its surface today. Internal forces created raised areas on the surface, such as the Tharsis Bulge, and stretched and split the surface to create rift valleys, such as the vast Valles Marineris (see pp.158–159). Landslides, wind, and water have since modified the rift valleys. Volcanic activity dates back billions of years and persisted for much of Mars's history. The planet may still be volcanically active today, although no such activity is expected. Lava eruptions of the past formed today's giant volcanoes, including Olympus Mons (see p.157).


OLYMPUS MONS
This mosaic of images of Olympus Mons taken by Viking 1 in 1978 looks deceptively flat—the volcano stands 15 miles (24 km) above the surrounding plain.


VALLES MARINERIS
The Valles Marineris is a complex system of canyons that cuts across Mars at an average depth of 5 miles (8 km). If it was on Earth, it would stretch across North America.



MARS MAPS
These four views combine to show the complete surface of Mars. They have been labeled to show large-scale features, as well as the landing sites of some of the spacecraft sent to explore its surface.